

Alberta Home Insurance Premium Pressure Monitor

SQL, Python, and Power BI Analytics Capstone

Business Question

Is Alberta home insurance premium growth unusual enough to warrant deeper affordability, market, or catastrophe-risk review?

Stakeholder:

Provincial public policy / insurance affordability analyst

Key framing:

This analysis is a review-trigger tool, not a causal proof model.

Data and Analytics Workflow

Data sources:

- Alberta home insurance CPI
- Alberta all-items CPI
- Canada home insurance CPI
- Residential construction-cost context
- Curated catastrophe events
- Alberta auto insurance CPI sensitivity check

Workflow:

Raw data → SQLite warehouse → SQL views → Python analysis → Power BI dashboard

Purpose:

- Show technical process without overloading the audience.

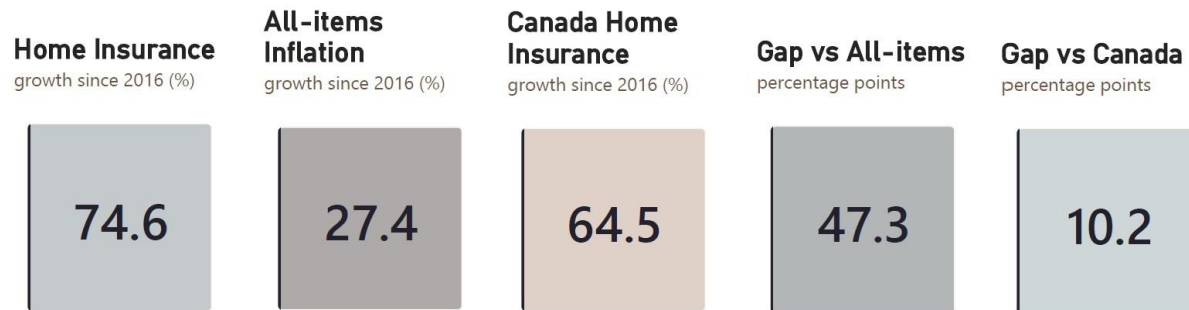
Alberta Home Insurance CPI Rose Faster Than Key Benchmarks

- Alberta home insurance CPI: **+74.6%**
- Alberta all-items CPI: **+27.4%**
- Canada home insurance CPI: **+64.5%**

Alberta home insurance CPI rose much faster than general inflation and exceeded Canada home insurance growth by 2025.

Alberta Home Insurance Premium Pressure Monitor

Tracking Alberta home insurance CPI against inflation and national insurance benchmarks

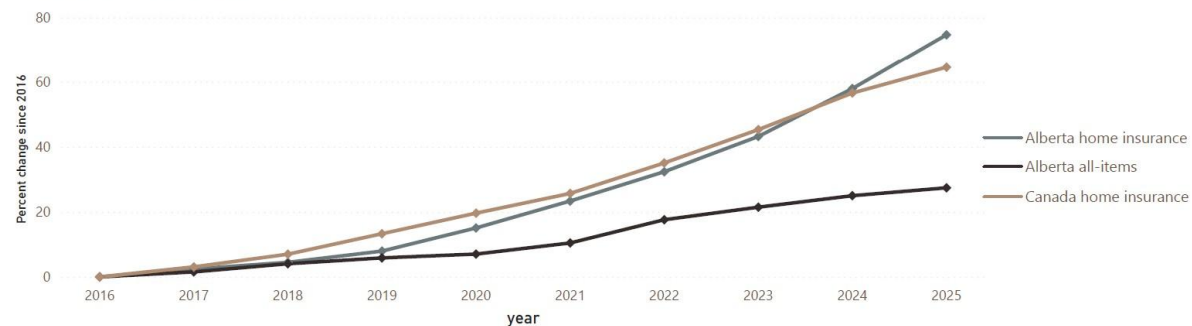


Review Signal

- Further review may be warranted.

CPI Growth Since 2016: Alberta Home Insurance vs Benchmarks

Indexed annual growth; 2016 = 0%



Alberta home insurance CPI rose much faster than Alberta all-items CPI from 2016 to 2025 and exceeded Canada home insurance CPI growth by 2025.

This suggests the period may warrant deeper affordability, market, or catastrophe-risk review.

Dashboard (Pg.1)

Executive Review

- KPI cards summarize benchmark divergence.
- Main line chart compares Alberta home insurance against inflation and national insurance trends.
- Review signal indicates further review may be warranted.
- Wording avoids causal over claiming.

Alberta Home Insurance Premium Pressure Monitor

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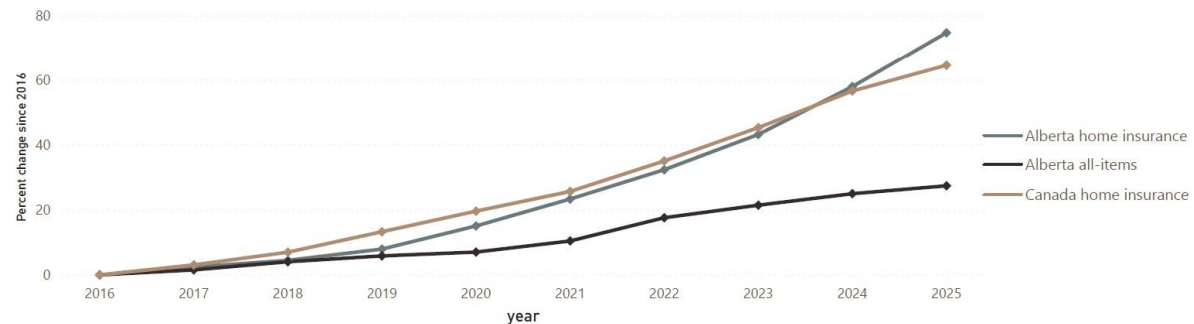


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Dashboard (Pg.2)

Context & Sensitivity

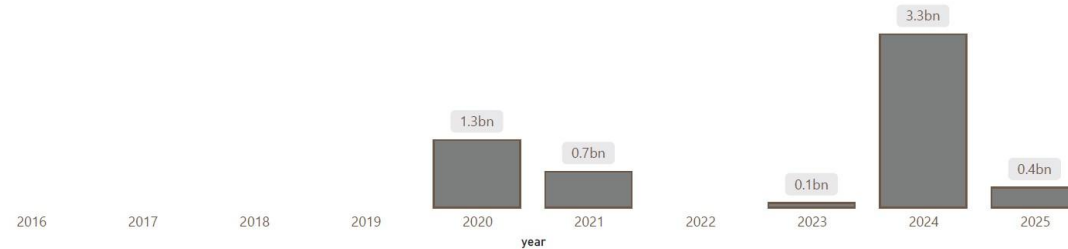
- Catastrophe losses cluster more heavily in later years.
- Auto insurance CPI sensitivity suggests some broader insurance-market pressure.
- Home insurance still exceeded auto insurance growth by 2025.
- Context supports review, but does not prove causality.

Context & Sensitivity Review

Catastrophe losses, method caveats, and insurance-line sensitivity

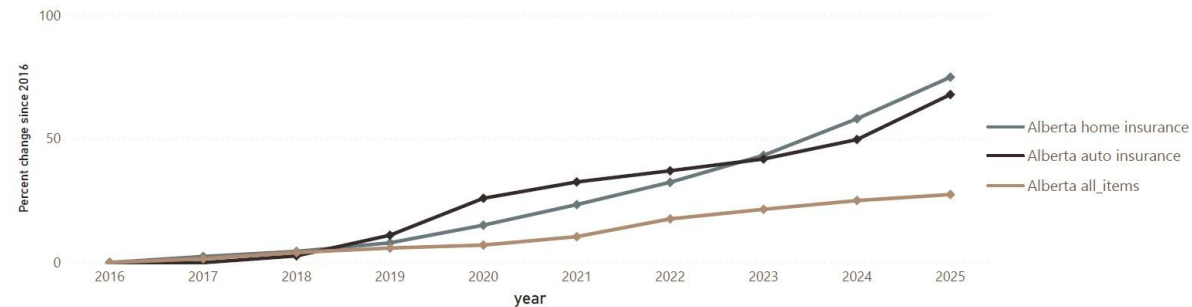
Curated Alberta Catastrophe Losses by Year in Billions

Curated major insured-loss events only; not an exhaustive catastrophe dataset



Sensitivity Check: Home vs Auto Insurance CPI Growth

Auto insurance helps test broader insurance-market pressure; it is not a property benchmark



Method Caveats

This dashboard is descriptive and exploratory.

Catastrophe events are timing/context markers, not causal proof.

Construction-cost comparison uses a 2017 baseline because the selected construction benchmark begins in 2017.

Auto insurance CPI is included only as a sensitivity check because auto and home insurance have different pricing drivers.

Dashboard (Pg.2)

Limitations

- Descriptive and exploratory analysis
- Catastrophe events are contextual markers, not causal proof
- Construction-cost benchmark uses a 2017 baseline
- CPI data are aggregated and do not show individual policy-level premiums
- Auto insurance CPI is a sensitivity check, not a property benchmark

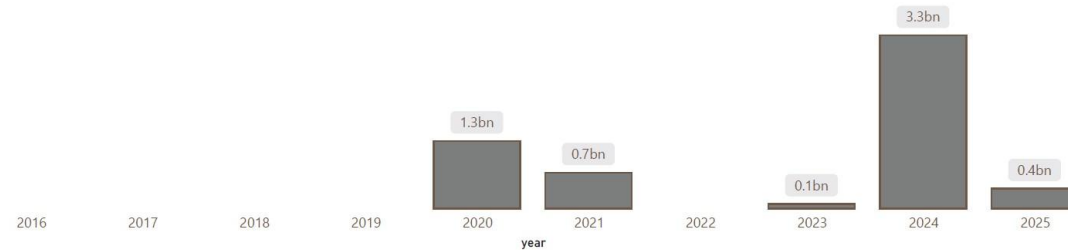
Limitations define the valid use case: this dashboard supports review and follow-up investigation, not causal attribution.

Context & Sensitivity Review

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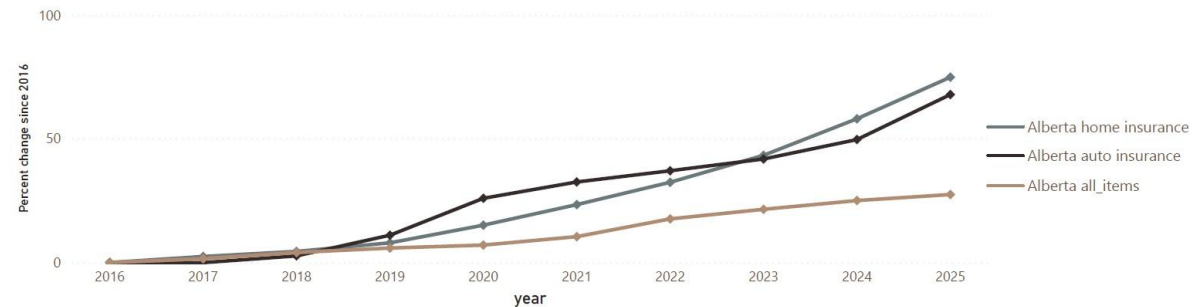
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Closing Message

The project identifies when Alberta home insurance CPI diverges from relevant benchmarks and helps stakeholders decide where deeper review is warranted

Recommendation

- Use benchmark divergence as a monitoring signal for affordability, market, and catastrophe-risk review.

Future Work

- Compare Alberta against other provinces
- Add more complete catastrophe-event data
- Investigate property-specific drivers such as claims severity, reinsurance, deductibles, and regional risk territories
- Explore GIS or regional exposure analysis as a separate project